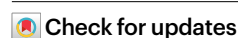


Benchmarking alignment methods for spatial transcriptomics data

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Reconstructing the three-dimensional molecular architecture of tissues from two-dimensional spatial transcriptomics slices is a central goal in spatial biology. Spatial alignment, the computational registration of multiple tissue slices using their spatial coordinates and gene expression profiles, provides the foundational framework for this integrative perspective. Although numerous alignment methods have emerged, a comprehensive benchmark to guide their application has been notably absent. Here we address this by systematically evaluating a diverse suite of leading methods. Executing 295 distinct alignment tasks across diverse datasets and technologies, our framework quantifies method accuracy, efficiency, usability and robustness, while also assessing the downstream impact of alignment quality. Crucially, our study systematically investigates performance in challenging real-world scenarios, uncovering substantial limitations in current tools. To address these bottlenecks, we propose and validate effective mitigation strategies. Finally, we provide practical guidelines to assist researchers in selecting the optimal alignment method and optimizing their analytical workflows.

Spatial transcriptomics (ST) technologies simultaneously capture gene expression profiles and spatial coordinates, enabling in situ gene expression characterization within the native spatial context of tissues. Current ST platforms primarily fall into two major categories^{1–3}: sequencing-based approaches (for instance, Visium⁴, Slide-seq⁵ and Stereo-seq⁶) and imaging-based approaches (for instance, MERFISH⁷, seqFISH⁸, STARmap⁹ and Xenium¹⁰). Despite their technical diversity, nearly all ST technologies operate on two-dimensional (2D) tissue slices, which constrains the information captured given the three-dimensional (3D) nature of biological tissues. The spatial alignment of these 2D slices represents a critical computational challenge that brings data

from multiple tissue sections into a unified coordinate system. This coregistration process enables cross-slice integration, providing a more comprehensive view of tissue molecular organization and facilitating downstream analyses that benefit from broader spatial context across multiple sections^{11–13}.

ST alignment can draw valuable insights from the well-established field of medical image registration^{14,15}. However, histology-based registration methods often fail in ST data when paired images are unavailable, staining protocols differ or manual landmarks are required. Moreover, ST data present unique challenges, such as sparse gene expression patterns, varying spatial resolutions across technologies

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and the need for cross-platform compatibility. These limitations have driven the development of numerous alignment methods that directly utilize expression matrices and spatial coordinates^{16–26}.

Given these ongoing challenges, spatial alignment remains challenging owing to tissue rotation, spatial warping and biological heterogeneity across slices¹⁹. Although one study has benchmarked alignment methods²⁷, it treated alignment as one of several tasks rather than providing dedicated, comprehensive evaluation and did not include several new methods. This existing benchmark study lacks systematic assessment of real-world scenarios such as tissue rotation and cross-technology alignment and fails to incorporate the latest ST technologies such as Xenium¹⁰ and CosMx²⁸, limiting its ability to guide method selection and development for practical applications. A comprehensive benchmark specifically focused on spatial alignment is essential.

To address this gap, we developed a comprehensive benchmark evaluating 11 spatial alignment methods across datasets from diverse ST platforms. For each method, a total of 295 tasks were executed, including 260 pairs of real slices and 35 pairs of simulated slices to be aligned. Using our unified evaluation framework with multidimensional quantitative metrics, we identified method-specific advantages and broadly generalizable approaches. Robustness assessments with systematic perturbations distinguished stable methods from those sensitive to conditions, whereas downstream task evaluation confirmed that high-quality alignment enhances analytical outcomes. Owing to the rapid development of ST technologies, new datasets are accumulating at an accelerating pace. For this reason, we offered a platform to accommodate these advancements, allowing users to apply our framework and code directly to benchmark methods with new data.

In addition, we explored three challenging scenarios critical for real-world applications but absent from previous benchmark studies—consecutive multiple slices, cross-platform alignment and large-scale datasets—revealing substantial methodological limitations including reduced accuracy and substantial memory consumption. To address these computational bottlenecks, we proposed three mitigation strategies that can be applied individually or in combination. Finally, we provided an executable user guide for optimal method selection based on dataset characteristics. This work establishes a systematic evaluation framework that highlights current challenges and provides a foundation for future spatial alignment algorithm development.

Results

Benchmarking framework for spatial alignment methods

Here we compared a total of 11 spatial alignment methods, including PASTE¹⁶, PASTE2¹⁷, STAligner¹⁸, GPSA¹⁹, SLAT²⁰, STalign²¹, CAST²², STAIR²³, SPACEL²⁴, Spateo²⁵ and SANTO²⁶. These methods predominantly utilize the gene expression matrix and spatial coordinates as input, performing linear, nonlinear or hybrid alignment and return transformed spatial coordinates as output (Fig. 1a). Notable exceptions include STalign, which requires single-cell resolution spatial transcriptomic data and only uses cell centroid coordinates, and SLAT, which exclusively returns matching point pairs that require additional processing for subsequent 3D reconstruction. Moreover, most methods support multislices (≥ 3 slices) alignment, whereas STalign, CAST and SANTO were restricted to pairwise alignment. To preclude the introduction of extraneous variables, we implemented a uniform strategy for alignment before stacking all slices into a common coordinate system (Methods).

To evaluate the performance of the alignment methods, we collected and constructed comprehensive benchmarking datasets comprising 240 slices from published studies and public database (Methods; Fig. 1c). These datasets were generated by a variety of spatial technologies, including 10x Visium⁴, 10x Visium HD⁴, ST²⁹, MERFISH⁷, Stereo-Seq⁶, BaristaSeq³⁰, STARmap⁹, STARmap PLUS³¹, Slide-seq⁵, Slide-seqV2³², Open-ST³³, Xenium⁴, Xenium 5K⁴ and CosMx³⁴. Critical data characteristics were considered to ensure the datasets could

represent diverse computational and biological scenarios, such as dataset scale, spatial resolution, tissue origin, slices overlap ratio, initial rotational offsets, warping level, interslice gene expression similarity and intraslice heterogeneity. On the basis of these real-world data slices, we further simulated 35 pairs of slices, yielding a total of 295 tasks to be executed for each method. All datasets were standardized into a unified format for method inputs (Methods).

After collecting the datasets, we developed a benchmarking framework across six key categories (Fig. 1b): (1) alignment accuracy, measured using gene-based metrics and landmark-based scores (for datasets with ground truth 3D coordinates, quantitative metrics such as the mean absolute error (MAE) were also calculated³⁵ (Methods)); (2) computational efficiency, quantified by runtime and peak memory usage; (3) robustness, evaluated by applying perturbations (for instance, varying slices overlap ratio and initial rotation angle); (4) downstream impact, assessed by measuring effects on 3D spatial clustering following alignment; (5) challenging cases, in which we tested each method on consecutive serial slices, cross-platform datasets and large-scale datasets (we also proposed solutions for the limitations encountered); and (6) usability, scored via a standardized usability scale (see Methods and Source data for more details). Finally, we provided practical guidance for users. Information on all datasets used, evaluation methods and metrics is summarized in Supplementary Table 1.

Method performance in conventional scenarios

We initially evaluated the performance of methods in conventional scenarios using small-scale, single-technology datasets, before moving on to more challenging cases. As these conventional scenarios are typically covered in methods papers, we first used them here to demonstrate the pipeline of our benchmarking framework. To do so, we first used datasets generated by two widely used spatial technologies—10x Visium⁴ and MERFISH⁷—as examples. Beyond visual inspection, we used quantitative metrics to assess alignment accuracy. On the basis of the biological hypothesis that corresponding niches from adjacent slices are more likely to have similar gene expression patterns and cellular composition, we developed gene-based metrics and landmark-based scoring, where higher values indicated greater spatial correspondence and thus superior alignment performance (Methods). For gene-based scoring, we defined overlapping regions between aligned slices, discretized them into grids and calculated the similarity of gene expression at corresponding positions after alignment using Pearson correlation coefficient (PCC), cosine similarity (Cos_Sim), structural similarity index (SSIM) and mutual information (MI). Higher similarity indicates better alignment performance, as it reflects greater consistency in gene expression within corresponding grid regions across slices. Concurrently, using region labels, we computed landmark-based scores through two measures: (1) the fraction of matching spots or cells pairs that have the same region labels post alignment and (2) the overlap ratio between the 2D geometries derived from the corresponding regional boundaries across aligned slices. These complementary metrics were synthesized into an overall ranking score for holistic method comparison (Methods).

For the next-generation sequencing (NGS)-based 10x Visium platform, we evaluated the method performance using the human dorsolateral prefrontal cortex (DLPFC) dataset as an example³⁶. This dataset includes three individuals, each containing four slices. The first and last pair of slices were located 10 μm apart, whereas the middle pair was 300 μm apart (Methods). Our evaluation focused on individual III (Data1–4), where each slice contained an average of 3,561 spatial spots profiled across 33,538 genes. We applied randomized rigid transformations on the slices and performed alignment using all methods. Methods that failed to run owing to memory overflow were excluded from this presentation. According to the overall ranking score (Methods; Fig. 2a, right-most column, higher ranks indicate better comprehensive performance), the top-performing methods

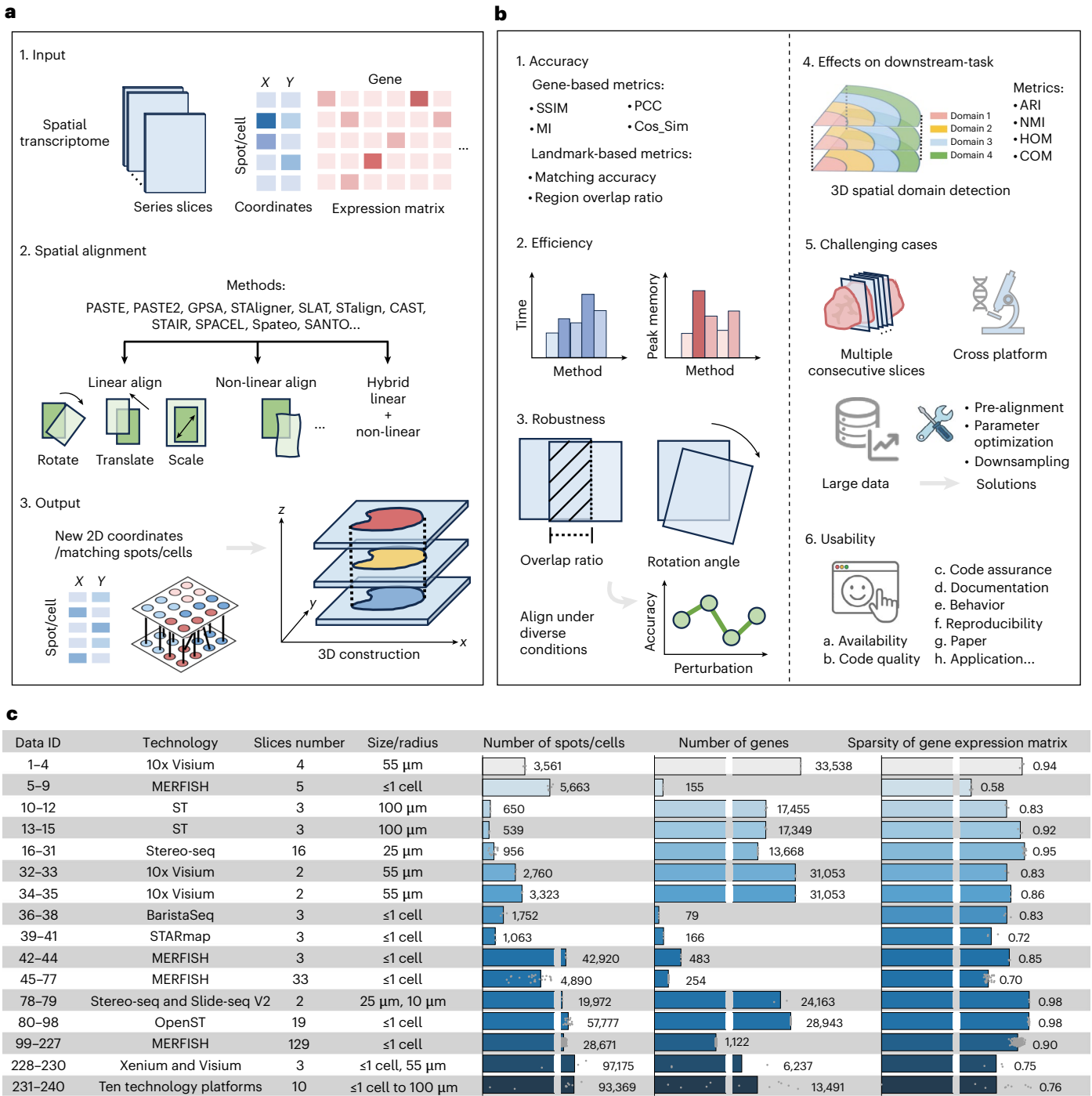


Fig. 1 | Overview of benchmarking workflow, methods and datasets.

a, A schematic illustration of the spatial alignment. The input of the task is ST data, such as spatial coordinates, expression matrices and other supplementary information (if required). Various methods are applied for alignment, which can be categorized into linear (involving rotation, translation and scale), nonlinear and hybrid approaches. The output of the task is typically a set of transformed new 2D coordinates or matching point pairs. Ultimately, all slices are stacked into a unified coordinate system to reconstruct the biological structure.

b, An overview of the evaluation categories and metrics. There are six main components. Accuracy was assessed using a combination of gene-based and landmark-based metrics (Methods). The efficiency evaluation recorded the runtime and peak memory usage of each method. For robustness, we designed

two perturbation conditions: varying degrees of overlap and different initial rotation angles, under which the performance of each method was examined. In the downstream analysis, we used 3D spatial clustering as an example to investigate how alignment influences subsequent downstream tasks. For challenging scenarios, we considered three representative cases and proposed potential solutions for handling large-scale datasets. We also assessed the usability of each method using a standardized evaluation scale. **c**, An overview of datasets used in this study, including data identifier (ID), technology platform, number of slices, spatial resolution, number of spots/cells, number of genes and data sparsity. Bar lengths represent mean values for all slices, with exact numbers labeled. The number of scatter points corresponds to the number of slices.

(Fig. 2a, right-most column, highlighted in red square) for the DLPFC dataset were PASTE2, SPACEL, STAIR and CAST, consistent with visual inspection results (Supplementary Fig. 1a). Further analysis revealed that PASTE2, STAIR and SPACEL maintained optimal performance in other 10x Visium datasets (Supplementary Fig. 2a,b).

Regarding MERFISH, an imaging-based spatial transcriptomic technology, we evaluated performance using the mouse hypothalamic preoptic region dataset³⁷ (Data5–9). This dataset comprises five slices, each slice containing an average of 5,663 cells and 155 genes (Methods). In this dataset, Spateo (using either rigid or nonrigid alignment) and PASTE2 demonstrated superior performance both visually (Supplementary Fig. 1b) and quantitatively (Fig. 2b), a finding that diverged from the results observed in Visium data. Further validation on supplementary datasets (Supplementary Fig. 3) confirmed Spateo's consistent outperformance across MERFISH datasets. STAIR also exhibited competitive accuracy, although its effectiveness was predominantly observed in datasets with a limited number of slices.

We calculated Kendall's tau correlation coefficients (Methods) for rankings obtained by sorting the metric values accordingly and observed strong agreement among the most metrics (coefficients typically >0.7), indicating high reliability of the rankings. A minority of correlations, however, fell below 0.5, suggesting complementarity between certain metrics (Fig. 2c,d).

Impact assessment on downstream task

We next examined how different alignment methods affected the performance of downstream task. To assess this, we respectively used the original slices and slices aligned by various methods as input for 3D spatial clustering in the DLPFC dataset with GraphST¹³, a graph self-supervised contrastive learning method that requires alignment before joint spatial clustering. The clustering performance was evaluated using the adjusted Rand index (ARI), normalized MI (NMI), the homogeneity score (HOM), the completeness score (COM) and an overall score combining all metrics (Methods). Encouragingly, most alignment methods improved the downstream performance compared with using unaligned slices, with 12 out of 15 outperforming the control group (Fig. 2e). The top-performing alignment methods, such as SPACEL, CAST and PASTE2, achieved the best downstream task performance in 3D spatial domain identification as expected, with all metrics showing approximately 0.3 value improvements over the baseline (Fig. 2e). Overall, these findings indicated that high-quality alignment generally led to improved accuracy in downstream 3D spatial domain identification. Consequently, in workflows where spatial reconstruction serves as an upstream step, alignment methods should be rigorously compared and selected to optimize the outcomes of downstream analysis.

Robustness assessment of alignment methods

To assess the robustness of methods, we simulated two perturbation conditions using real-world datasets (Methods): (1) varying overlap ratios between slices, where the query slice was cropped at specified proportions to generate datasets with different overlap ratios and then aligned to the reference slice (Fig. 3a) and (2) different angular deviations, where the query slice was rotated in 15° increments (0–360°) to create slices with varying initial angular offsets for alignment (Fig. 3b). In addition to the demonstrated Visium and MERFISH datasets, four additional datasets from distinct technologies were included in the evaluation (Supplementary Fig. 4a–d). The alignment performance was quantified using the established accuracy metric.

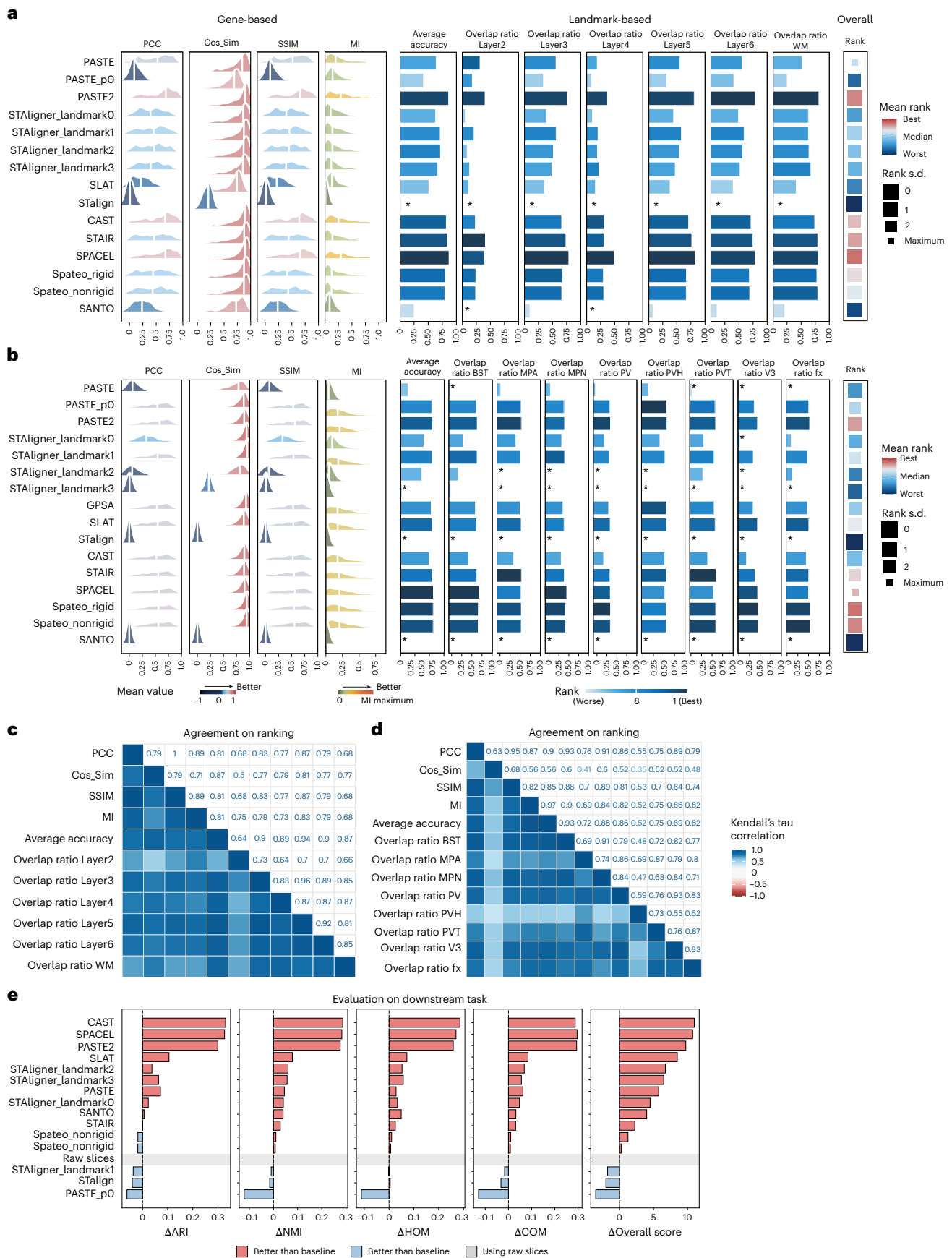
For condition (1), we observed that the accuracy of most methods increased with higher overlap ratios, consistent with previous studies²⁷. SPACEL showed exceptional stability, maintaining high accuracy across various conditions (Fig. 3e,f and Supplementary Fig. 4a–d). Regarding the perturbation condition (2), PASTE2, STAIR, SPACEL, Spateo, SANTO and STAligner (using optimal landmark) showed robust performance across all datasets. By contrast, PASTE and SLAT exhibited angular sensitivity, indicating that their alignment accuracy is heavily dependent on initial angular configuration (Fig. 3c,d and Supplementary Fig. 4a–d). We further found that slice pairs with angular deviations within ±30° generally yielded better alignment results. Considering both conditions, PASTE2 and SPACEL demonstrated stable performance with optimal effectiveness (as indicated in Fig. 3g,h, the larger circles with more purple color represent methods with higher accuracy and greater stability), whereas Spateo also exhibited relatively stable results (Fig. 3g,h and Supplementary Fig. 4a–d). These findings are in line with method design principle because PASTE2 was specifically implemented to address issues such as partial overlap between slices¹⁷, and SPACEL utilized additional region information for alignment²⁴.

Overall performance under conventional scenarios and usability

Beyond the two example technologies that we analyzed before, we categorized the datasets that are compatible with most methods into two groups^{1–3} on the basis of their spatial transcriptomic platform to evaluate the overall performance in conventional scenarios. For NGS-based technologies, we evaluated five datasets from ST, Visium and Stereo-seq, comprising a total of 27 slices (Data1–4, Data10–12, Data16–31, Data32–33 and Data34–35). For imaging-based technologies, we evaluated four datasets from MERFISH, BaristaSeq and STAR-map, comprising 44 slices in total (Data5–9, Data34–35, Data39–41 and Data45–77). Following our established benchmarking framework, we comprehensively aggregated method performance across gene-based metrics, landmark-based metrics, runtime efficiency and peak memory

Fig. 2 | Evaluation in conventional scenarios and impact on downstream task. **a**, The quantitative evaluation on the Visium DLPFC dataset (Data1–4). For gene-based metrics—PCC, Cos_Sim, SSIM and MI—kernel density curves of marker genes are shown with white lines indicating the median and color representing the mean value. For landmark-based metrics (accuracy and overlap ratio of corresponding regions), bar plots are shown. The accuracy is the proportion of matching point pairs that belong to the same region after alignment. The overlap ratio between major corresponding regions in aligned slices was calculated, with each ratio recorded as a metric named 'overlap_ratio_region'. The bar height represents the metric value (the asterisk denotes a value <0.05), and the color represents the rank. For all these metrics, higher values indicate better performance. The overall rank was obtained by averaging the rankings of all metrics (Methods); the color denotes the ranking order (methods with higher ranks are represented with redder colors), and shape size represents the standard deviation of rankings—larger shapes indicate a lower dispersion of metric rankings and thus more reliable overall rankings. For STAligner, we executed alignment using each landmark identified by STAligner and recorded individual outcomes. The results were labeled as

STAligner_landmarkN (N = 0,1, ...) accordingly. **b**, The quantitative evaluation on the MERFISH dataset of the mouse hypothalamic preoptic region (Data5–9); same as **a**. **c**, The consistency of metrics on the DLPFC dataset. Kendall's tau correlation is shown between ranking vectors, where each vector was generated by sorting all methods on the basis of the specific metric. Bottom left: a heat map of correlation values. Top right: numeric correlation values. A value closer to 1 indicates higher consistency between the rankings. **d**, The consistency of metrics on the MERFISH mouse hypothalamic preoptic region dataset; same as **c**. **e**, The downstream task results for all methods on the DLPFC dataset using GraphST. Bar plots show the performance difference compared with the control group across various metrics. The control group is highlighted with a gray background; the red bars indicate better performance than baseline, and the blue bars indicate worse performance. The methods are ordered by overall score (the last column). WM, white matter; BST, bed nuclei of the strata terminalis; MPA, medial preoptic area; MPN, medial preoptic nucleus; PV, periventricular hypothalamic nucleus; PVH, paraventricular hypothalamic nucleus; PVT, paraventricular nucleus of the thalamus; V3, third ventricle; fx, columns of the fornix.



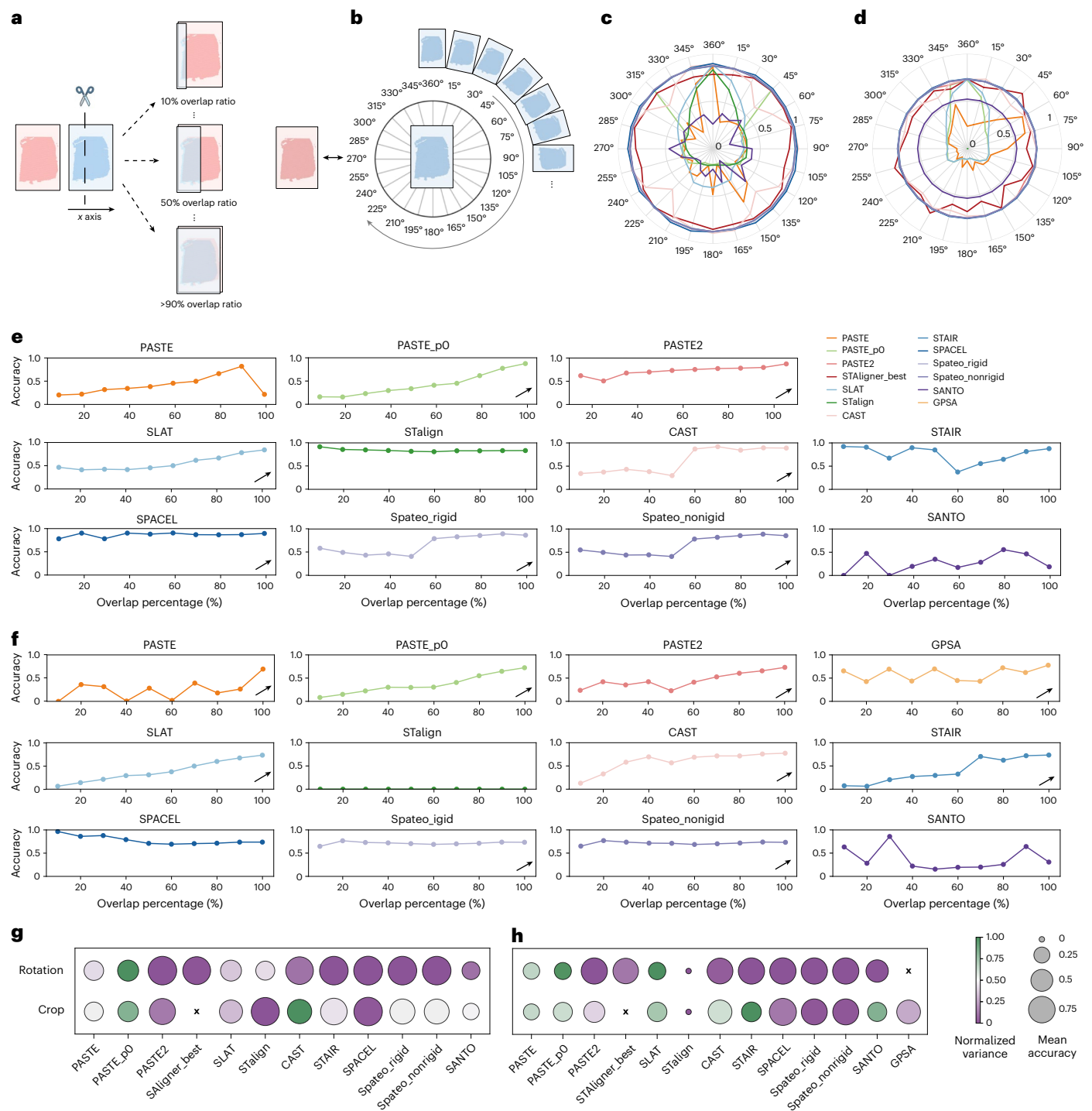


Fig. 3 | Robustness evaluation under perturbations. a, A schematic illustration of how cropping slices alters the overlap ratio between slices. Partial overlaps were simulated by shifting the cropping window along the x axis of the target slice in 10% width increments, extracting the left portion parallel to the y axis. This generated slices with -10–100% overlap, which were aligned to the reference to assess performance across overlap levels. **b**, A schematic illustration of how rotating slices changes the initial spatial offset between slices. The target slice was rotated clockwise around its centroid in 15° steps, spanning the full 0–360° range. Each rotated slice was aligned to the reference slice to assess the method's robustness to angular variation. **c, d**, Radar plots showing the alignment accuracy under varying initial rotation angles on the DLPFC (**c**) and

mouse hypothalamic preoptic region (**d**) datasets. Each axis corresponds to a rotation angle. Colors represent different methods, with the legend showing the method–color correspondence below. **e, f**, Line plots show accuracy under varying overlap ratios on the DLPFC (**e**) and mouse hypothalamic preoptic region (**f**) datasets. Each subplot represents a method; the x axis denotes the overlap ratio (10–100%), and the y axis shows the corresponding accuracy. Arrows indicate the overall trend direction (an increasing trend). The legend shows the method–color mapping. **g, h**, The mean and variance of accuracy under the two perturbation settings on the DLPFC (**g**) and mouse hypothalamic preoptic region (**h**) datasets. Higher mean and lower variance indicate better performance and robustness, as illustrated by the large purple circle in the figure.

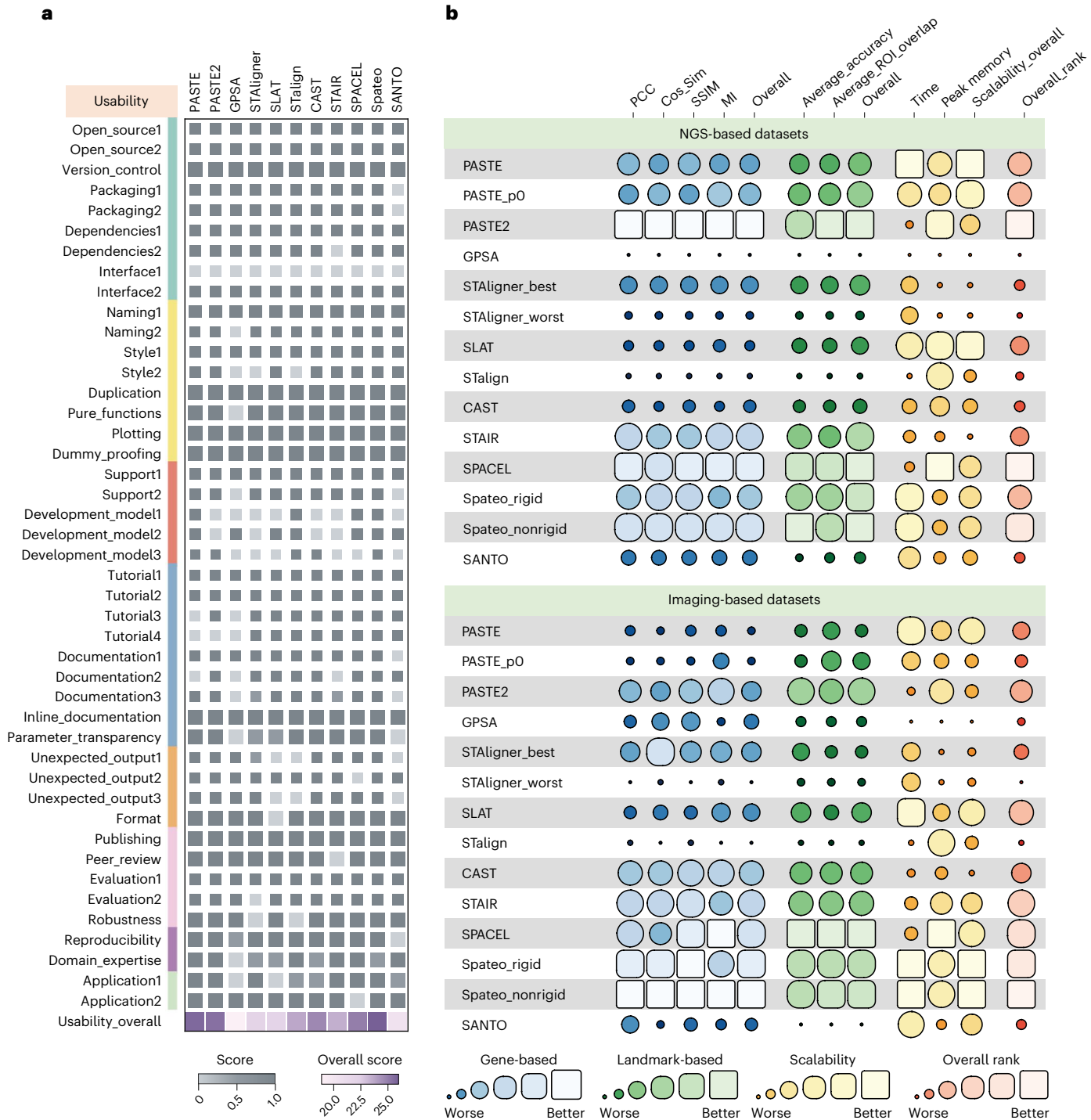


Fig. 4 | Summary of overall performance and usability. **a**, The usability scoring. Scores are visualized through a color gradient. For each item, a score closer to 1 is considered better usability. The box size indicates the weight of each component, and the color on the left represents the corresponding category: availability, code quality, code assurance, documentation, behavior, reproducibility, paper and application. Please refer to Supplementary Table 2 for detailed information regarding the meanings of each item and their respective weight proportions. Bottom row: the overall usability score was computed as a weighted sum of all items. A deeper purple color indicates a higher overall score, reflecting better usability. **b**, An overall performance summary. Datasets are grouped into NGS-based and imaging-based categories. The NGS-based category

includes multiple datasets from three different technologies (ST, Visium and Stereo-seq), comprising a total of 27 slices. The imaging-based category also consists of multiple datasets from three technologies (MERFISH, BaristaSeq and STARmap), totaling 44 slices (Methods). Evaluation metrics were categorized into three groups: gene-based, landmark-based and scalability. For each metric, method rankings were computed as previously described and averaged across all datasets. To summarize the performance within each group, the rankings of individual metrics were averaged, shown in the group's final column. An overall ranking score was then derived by integrating all metrics (last column; see Methods for details). In the visualization, better performance is indicated by lighter colors and shapes closer to squares.

consumption. All methods were ranked within each category and then synthesized into an overall score (Methods).

According to the overall ranking (Fig. 4b), PASTE2 and SPACEL demonstrated the top performance not only on Visium data (Fig. 2a and Supplementary Fig. 2) but also across all NGS-based datasets, ranking top three alongside Spateo. This preeminence, however, came at the cost of efficiency compromise: PASTE2 and SPACEL required longer computation time, and Spateo consumed a little more memory. Despite slightly inferior efficiency, such a trade-off would be acceptable if users primarily prioritize alignment accuracy. For imaging-based datasets, Spateo (using either rigid or nonrigid alignment) attains the highest overall scores, with STAIR and SPACEL also performing well.

It is important to note that the composite ranking reflected general performance rather than universal applicability across all scenarios. For instance, STalign specializes in single-cell resolution data and performs optimally on slices with distinct biological structures, and the efficacy of STaligner partially depends on landmark selection, requiring user-guided inspection. As for SLAT, we additionally applied simple rigid transformations for standardized benchmarking, although this step becomes unnecessary if users solely require matching spots for biological analysis. In specific scenarios, certain methods may yield better outcomes depending on dataset characteristics, necessitating user-driven testing and comparative validation.

To further quantitatively investigate these similarities and differences across datasets, we computed the rank correlations of method performance orderings across different datasets (Supplementary Fig. 12). A clear separation was observed between sequencing-based datasets and imaging-based datasets, with substantially higher within-group correlations and highly consistent method rankings. Nevertheless, two datasets exhibited notable exceptions. We attribute these deviations to multiple potential factors, including differences in initial slice positioning, morphological characteristics, tissue complexity, as well as the incorporation of runtime and memory usage into our overall ranking metric. Despite these variations, pairwise correlations across nearly all datasets remained above 0.5. This indicates that, although dataset-specific characteristics do exist, the overall relative performance ordering of the methods is more consistent than initially anticipated.

We further adapted a usability framework to systematically evaluate the usability of each method³⁸. The assessment criteria included availability, code quality, code assurance, documentation, behavior, reproducibility, paper and application. We assigned different weights to these categories for overall scoring. Information regarding the classification, detailed meanings, weights and other details for each item in the figure can be found in Supplementary Table 2. On the basis of the final results (Fig. 4a), Spateo and PASTE series ranked highest as the most user-friendly options. Other methods also maintained adequate usability, whereas GPSA and SANTO presented barriers, particularly for users with limited programming expertise.

Evaluation under challenging cases

For comprehensive benchmarking, we further extended our consideration to three biologically challenging scenarios: cross-platform

datasets, serial slices (more than ten slices) and large-scale data. The cross-platform alignment is a meaningful and realistic task but has been overlooked in previous research^{16,19,24,25}. For example, the Brain Initiative Cell Census Network project produced mouse brain datasets across multiple ST platforms^{31,39–41}. Aligning these datasets is necessary for integrating complementary information and mitigating the limitations of individual technologies. However, this task remains highly challenging. The difficulties of cross-platform alignment stem from the inherent feature heterogeneity across ST platforms, including unmatched spot or cell quantities, resolution and gene panel sizes. As for the second scenario, the alignment of multiple consecutive slices presents challenges in achieving global 3D reconstruction, with processing time typically increasing as the number of slices grows. Finally, the large-scale datasets are prone to encounter computational hardware constraints, which often culminate in out-of-memory errors that prevent successful execution.

For the first scenario, we collected two cross-platform datasets for evaluation. The first dataset comprised two mouse olfactory bulb ST slices profiled by Stereo-seq⁴² and Slide-seqV2³², respectively, both exhibiting well-preserved structures (Fig. 5a, Data78–79). In this dataset, the STaligner and CAST showed superior performance across all evaluation metrics, whereas SLAT also achieved competent results, consistent with visual assessments (Fig. 5b and Supplementary Fig. 5). The second dataset tests a more challenging case, consisting of slices from ten major ST platforms (including ST⁴³, 10x Visium⁴⁴, MERFISH⁴⁵, STARmap PLUS³¹, Stereo-seq⁴⁶, Slide-seq⁴¹, Xenium⁴⁴, Xenium 5k⁴⁴, Visium HD⁴⁴ and CosMx³⁴) (Fig. 5c, Data231–240). These datasets exhibited substantial technical variability in spot numbers and gene panel sizes, with spatial resolutions spanning subcellular to 100- μ m scales, despite all maintaining clear biological structures. We performed pairwise alignment across all ten slices, documenting both successful executions and alignment outcomes. Our analysis revealed that although several methods executed successfully, they generally exhibited poor performance (Fig. 5d and Supplementary Figs. 6 and 7). Only the alignments between MERFISH, Xenium and Xenium 5k achieved slightly satisfactory results, underscoring current deficiencies in cross-platform alignment and highlighting the need for novel algorithms tailored to this challenge.

For multiple consecutive slices analysis, we selected two datasets respectively derived from Stereo-seq⁴⁷ (16 slices with 400–1,300 cells per slice, Data16–31) and from Open-ST³³ (19 slices with 1 million cells, Data80–98). Alignment accuracy was quantified using MAE between aligned coordinates and ground truth (Methods). In addition, two MERFISH datasets (33 slices and 129 slices respectively, Data45–77, Data99–227) were included for benchmarking (Methods; Supplementary Fig. 3b,c). Among all methods capable of aligning these datasets, Spateo demonstrated consistent top performance, exhibiting distinct advantages in handling multi-consecutive-slice alignment (Fig. 5f,g and Supplementary Fig. 3b,c). Other methods either performed well only on specific datasets or failed to execute when confronted with large number of slices.

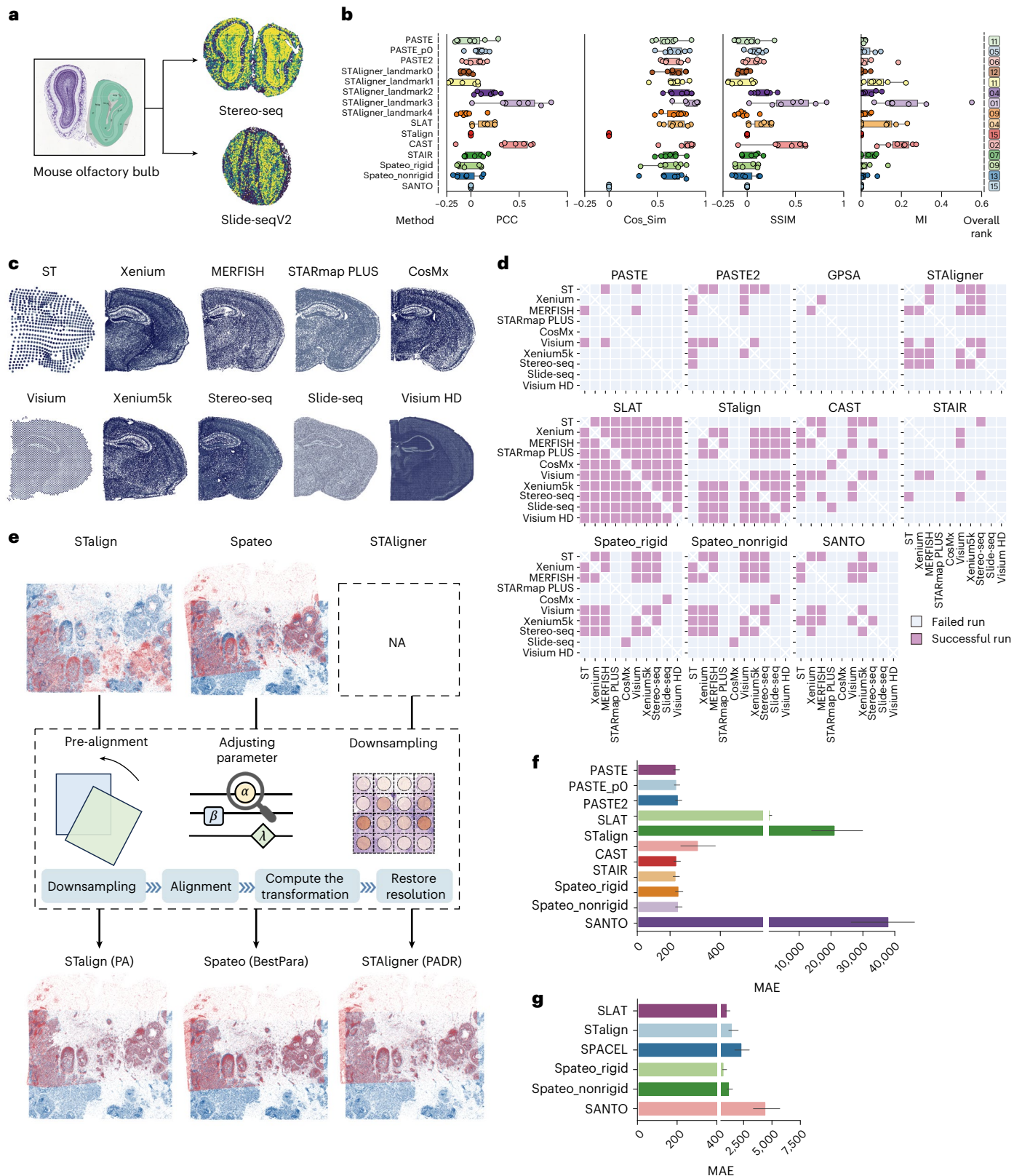
During evaluations across datasets, we observed that high-resolution and large-scale datasets generally trigger out-of-memory errors and execution failures. We examined two consecutive

Fig. 5 | Challenging scenarios: cross-platform, multiple consecutive slices and large-scale data. **a**, The cross-platform mouse olfactory bulb dataset visualization (Data78–79). **b**, Box plots of alignment metrics on Data78–79. Each box shows the distribution of metrics values for marker genes. Center line, median; box limits, interquartile range; whiskers, 1.5 $\times$ interquartile range. **c**, The visualization of the cross-platform mouse brain coronal sections dataset (Data231–240). Each uses a different color scale to enhance visibility of biological structures. **d**, The alignment results on Data231–240. Rows and columns represent slices from different platforms. The color means whether a method ran successfully for a given pair. A run is considered failed when memory overflow occurs or any error is reported by the method. No time restrictions were

imposed. Hardware specifications: 503 GB of system memory, 80 GB of GPU memory. **e**, Three solutions for large-scale Xenium human breast cancer data (Data229–230). Representative alignment results from methods after applying this strategy are shown. The blue path of the central flowchart illustrates the resolution-recovery strategy. **f,g**, The alignment performance evaluated by MAE on the Stereo-seq Flysta3D (**f**) and OpenST human metastatic lymph node (**g**) datasets (in **f**, Data16–31, $n = 16$; in **g**, Data80–98, $n = 19$). Each bar represents the average MAE. The 95% confidence interval is included for each line. A smaller MAE value reflects superior performance. Methods with failed runs are excluded from the display. PA, pre-alignment; BestPara, parameter optimization; DR, downsampling resolution; PADR, combination of PA and DR; NA, not applicable.

human breast cancer slices from Xenium as a case¹⁰ (Data229 and Data230), where only a few methods could complete alignment, and virtually none delivered satisfactory results (Supplementary Fig. 8a,b). To address these challenges, we propose three practical solutions (Methods; Fig. 5e and Supplementary Figs. 8c,d and 9). First, implement a coarse pre-alignment involving manual observation and rigid transformations to achieve approximate angular consistency before

formal fine alignment. This step can substantially reduce computational time while improving precision. For instance, STalign achieved near-perfect alignment with substantially reduced runtime following this preprocessing (Fig. 5e and Supplementary Fig. 8b,c). The second approach targets methods with well-documented tutorials of parameter selection, such as Spateo, where systematic parameter optimization enables the attainment of satisfactory alignment outcomes (Fig. 5e



and Supplementary Fig. 8b). The third strategy, universally applicable across all methods, primarily addresses out-of-memory and excessive runtime issues. The workflow involves: (1) reducing data resolution and aligning downsampled slices, (2) calculating transformations from the reduced-resolution alignment and (3) applying these transformations to the original high-resolution data for resolution recovery (Fig. 5e and Supplementary Figs. 8d and 9). Notably, this process performs iterative testing of different resolution levels to identify the optimum for maximal alignment accuracy, and we tested 25 different resolutions in our study. As evidenced by STAligner—which initially failed owing to memory limitations—this approach enabled successful alignment with high precision after strategy implementation. It is worth noting that a single strategy may not be universally applicable to all methods. Therefore, rather than relying on any one strategy alone, we also recommend that users combine multiple strategies on the basis of specific requirements to optimize alignment performance.

Comparisons with prior study and potential implications for next-generation methods

Overall, our benchmark has revealed several findings, which are described in detail in the preceding sections. We also compared this study with a previous benchmark study²⁷ and summarized the similarities and differences in Supplementary Note 1. In the following subsection, we attempt to preliminarily explore the factors underlying the observed performance differences across methods to offer several potential insights for the next-generation methods.

In conventional scenarios, we categorized the datasets into sequencing-based and imaging-based groups and observed that the top-performing methods differ substantially between these two categories. We think that the observed performance discrepancies can be attributed to four fundamental intrinsic data characteristics. First, the resolution and the cell-mixture effect substantially impact local specificity. Sequencing-based data are spot-based, with each spot typically aggregating transcripts from multiple cells, thereby blurring fine tissue boundaries. When alignment algorithms compute interslice similarity, this mixing effect reduces the specificity of local features, causing algorithms to favor global anatomical structure matching. By contrast, imaging-based platforms achieve single-cell resolution, facilitating the capture of subtle morphological variations but at the cost of higher computational complexity and sensitivity to segmentation noise. Second, data sparsity and dropout introduce varying levels of uncertainty. The severe zero-inflation characteristic of sequencing-based data degrades the accuracy of interslice similarity matrices for expression, whereas imaging data generally exhibit higher sensitivity and cleaner signal-to-noise ratios for detected genes. Third, the scope of the feature space dictates alignment robustness. Sequencing-based platforms benefit from whole-transcriptome coverage, which provides rich information, whereas imaging-based alignment relies on targeted panels and is therefore highly dependent on the inclusion of spatially discriminative marker genes to distinguish anatomical regions. Finally, spatial topology and grid regularity impose distinct geometric constraints. The rigid, fixed lattice of sequencing-based data provides strong spatial priors, contrasting with the unstructured, density-heterogeneous point clouds of imaging-based data that demand greater algorithmic robustness (for instance, graph neural network-based approaches).

In addition to the data-intrinsic differences arising from distinct technological platforms, we further sought to evaluate the respective contributions of transcriptional similarity, spatial distance and morphological similarity to the alignment task. We constructed three simulated datasets derived from real MERFISH data (Supplementary Fig. 10a, identifiers 5–9): (1) we set the gene expression profiles of all cells within each slice to be identical (averaged on the basis of total counts). In this case, alignment methods cannot learn any informative features from gene expression

(Supplementary Fig. 10b). (2) For all slices except the first one, we assigned all cells the exact same coordinates (centered at the slice center). Under this condition, alignment methods cannot extract any meaningful spatial coordinate features or relative positional relationships between cells and can only rely on gene expression similarity for alignment (Supplementary Fig. 10c). In both of the above scenarios, we added a very small amount of random noise to avoid numerical issues caused by complete identity. (3) The original data exhibit clear cell density variations that form distinct morphological patterns (Supplementary Fig. 10e, top). We rearranged all cells into a regular grid according to their sorted coordinates, thereby eliminating density differences (Supplementary Fig. 10d,e). This preserves relative positional relationships between cells but removes local morphological features arising from cell density heterogeneity (Supplementary Fig. 10e, bottom).

We tested all methods on the above three simulated scenarios, noting that SPACEL was excluded from this part of the evaluation for fairness as it requires region labels as input. In the first scenario, SLAT and PASTE_p0 still achieved reasonably good alignment performance (Supplementary Fig. 11a). This suggests that these two methods can effectively guide alignment using spatial coordinate information alone and do not strictly require gene expression information. For all other methods, however, gene expression information appears to be essential. In the second scenario, all methods completely failed to align the slices (Supplementary Fig. 11b). This demonstrates that relative positional or distance information between cells/spots is critical, and existing alignment methods cannot succeed using gene expression similarity alone. In the third scenario, many methods still achieved good global alignment (Supplementary Fig. 11c). This indicates that local morphological features (for instance, cell density heterogeneity) do not play a decisive role in global alignment for current alignment approaches, at least when sufficient gene expression information and clear spatial domains are present. Taken together, for existing alignment methods, the most critical factors remain the effective integration of gene expression information and relative spatial positional relationships, as the absence of either component leads to substantially degraded performance compared with the original data (Supplementary Fig. 2b). By contrast, local morphological features appear to play only a limited, nondecisive role for current approaches. Therefore, future methodological developments can proceed along two complementary directions. On the one hand, new algorithms may continue to follow the weighting preferences observed in existing methods by prioritizing gene expression similarity and relative spatial positional relationships in the primary objective function, while de-emphasizing local morphological features (for instance, cell density variation) during the initial alignment stage and incorporating them only in subsequent refinement steps. On the other hand, an equally intriguing and potentially transformative avenue would be to deliberately depart from these established preferences and design novel frameworks that exhibit markedly different reliance on the various information sources, which could potentially uncover complementary principles and further advance the field.

Discussion

In this study, we performed a comprehensive benchmark analysis of 11 spatial alignment methods using datasets generated from various ST technologies. Integrating all the findings in this study, we offered practical guidelines with method selection criteria and workflow to assist researchers in the alignment task (see Fig. 6 for details).

We encapsulated the core functionalities of the proposed framework into a practical toolkit with detailed tutorials to improve development and application efficiency. The toolkit supports the rapid computation of evaluation metrics and direct comparison with existing alignment methods, enabling effective method evaluation and integration. It also provides executable solutions for large-scale data,

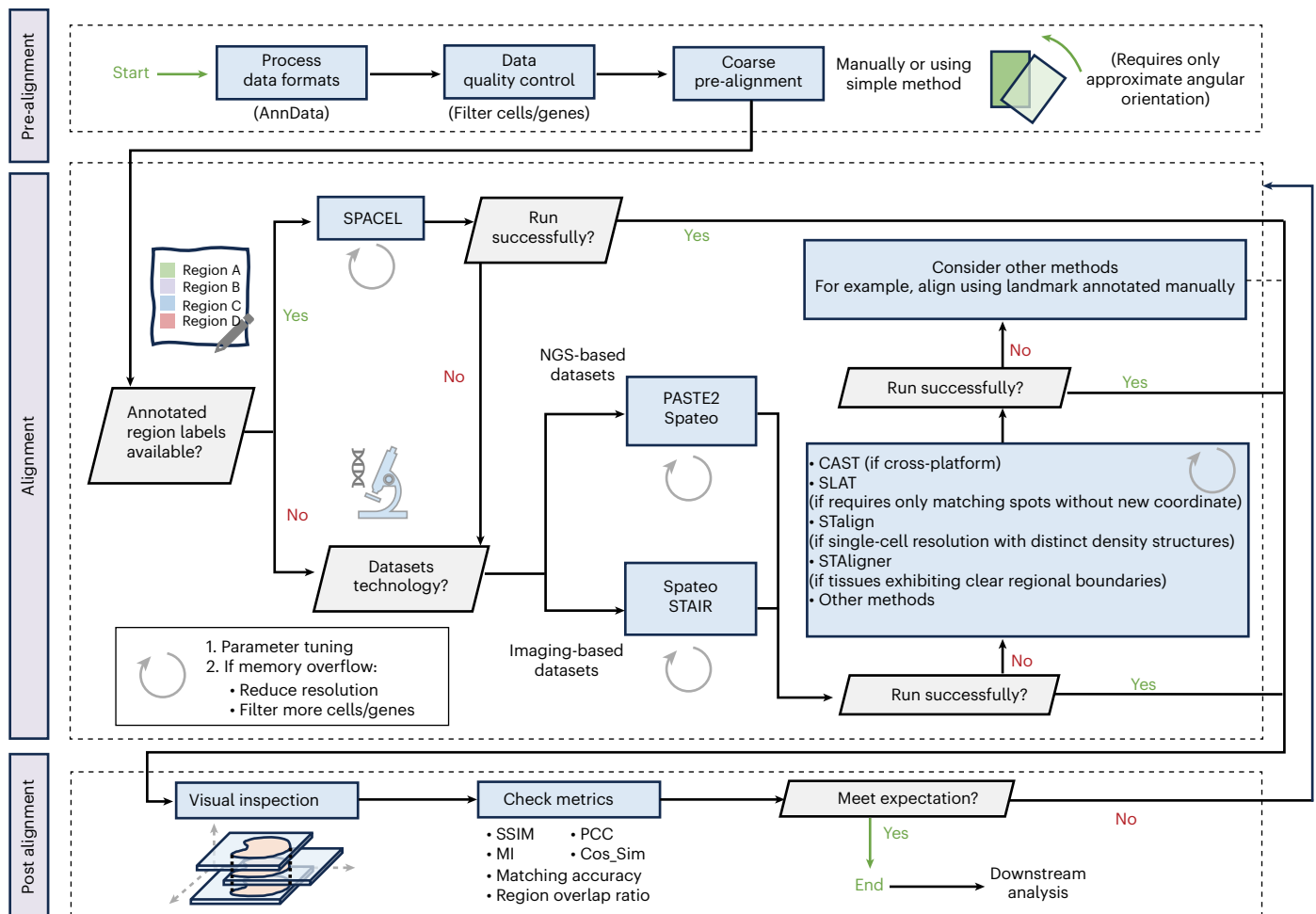


Fig. 6 | A practical guideline for alignment. A step-by-step guide for users to perform spatial alignment. The process includes three major stages. Before formal alignment, users are advised to process the data format, perform quality control and carry out pre-alignment steps (for example, correcting the initial slice orientation). During the formal alignment stage, methods should be tested sequentially according to the workflow. Each method may involve internal

iterations such as parameter tuning or applying troubleshooting strategies if out-of-memory errors occur; if the issue persists, proceed to the next method. After successful alignment, users should perform postalignment checks, including visual inspection and metric-based evaluation. If the alignment quality meets expectations, the downstream analysis can begin. Otherwise, return to the previous stage and continue testing alternative methods.

including interactive coarse alignment, grid-based downsampling, resolution restoration and method-specific parameter-tuning strategies for real-world applications.

On the basis of our work, in addition to the potential methodological insights mentioned in the final section, we believe that future developments should primarily focus on the following aspects. The first direction is the cross-platform alignment task, as demonstrated in the first challenging scenario. Although few existing methods explicitly show such alignment examples, this capability holds biological importance—for instance, aligning adjacent slices profiled by complementary technologies could compensate for platform-specific limitations and generate comprehensive analytical insights. Technology benchmarking studies could also leverage these aligned slices from different platforms for standardized regional comparisons. Unfortunately, we found that current methods still showed limited effectiveness in this cross-platform alignment, indicating an essential direction for further methodological development.

Alongside cross-platform alignment, cross-omics alignment for integrating multimodal data represented another challenging task worthy of exploration. Unlike several integration methods that embed multiomics information into a low-dimensional space, cross-omics alignment offers a more direct strategy by integrating multiomics

data within the physical 3D space, thereby restoring tissue landscape more faithfully. Several methods have demonstrated these capabilities, such as aligning ST data with images^{21,48} or epigenomic data²⁶. Although our study did not benchmark cross-omics tasks, this area remains highly promising. The accurate spatial alignment of diverse omics modalities—such as histological imaging, epigenomics, ST and spatial proteomics—could pave the way for comprehensive dynamic modeling across both spatial and time scale. For instance, aligning spatial omics data to temporally resolved medical imaging could facilitate spatiotemporal molecular profiling, providing valuable insights for clinical applications such as cancer outcome prediction, prognosis and response to neoadjuvant therapy.

In addition to methods focusing on spatial-level alignment, methods specifically designed for spatiotemporal omics have also emerged⁴⁹. Their core innovation lies in the ability to align data across different time scales. For example, such methods can accurately align ST data collected at different developmental stages (for instance, during embryogenesis), providing critical tools for deciphering the spatiotemporal regulatory mechanisms of dynamic developmental processes. However, regardless of the specific direction of development, we anticipate the new methods will commonly face a fundamental challenge—the limitations imposed by large-scale data.

As ST technologies advance, many platforms now achieve subcellular-level resolution, accompanied by exponential growth in data scale. In our study, we proposed three feasible solutions: preliminary coarse alignment, parameter optimization and data downscaling. Beyond the demonstrated approach of resolution reduction, users could also implement quality-control measures such as cells or genes filtering to resolve memory overflow and excessive runtime issues. However, despite existing solutions, no current method achieves both rapid and precise alignment for such large-scale datasets without auxiliary information (for instance, hematoxylin and eosin (H&E) images, cell type or region annotations) or additional preprocessing. Given the anticipated predominance of such large-scale data in future spatial transcriptomic studies, next-generation alignment methods should prioritize these computational constraints and alleviate this challenge.

Finally, given the remarkable reasoning, integration and generative capabilities that large language models (LLMs) have demonstrated in single-cell RNA sequencing^{50,51} and ST⁵², and with the recent study of foundation models for alignment tasks⁴⁸, we anticipate that LLMs will play a more important role in advancing the understanding of spatial tissue architecture—particularly in cross-modal alignment tasks. For instance, an existing foundation model, OmiCLIP⁴⁸, achieves integration of histopathology with ST data, enabling the cross-modal alignment of H&E image and ST data. It initially embeds patch-level transcriptomic profiles and H&E images into a latent space, followed by the application of an adapted coherent point drift algorithm to align the two embeddings, then accomplishing the alignment task. This model offers inspiration for further cross-omics alignment, such as extending to ST, spatial proteomics and spatial metabolomics. However, several challenges remain: (1) paired spatial multiomics data are still relatively scarce, and (2) determining how to appropriately encode various spatial multiomics data as input remains complex. For example, existing methods have represented the transcriptomic profile of a tissue patch as a ‘sentence’ composed of top-ranking highly expressed genes. Overall, the strengths of LLMs in contextual awareness, incorporation of prior knowledge and automation of analytical workflows will open up new possibilities for complex spatial tissue reconstruction, visualization and downstream analyses in 3D spatial biology. As these research directions continue to gain momentum, we foresee that an increasing number of datasets and methods will emerge in the future. This will not only enable more comprehensive and rigorous benchmarking but also drive deeper insights into the spatial and temporal organization of complex biological systems.

Methods

Benchmarked methods

The selection of methods was based on a combination of factors, such as citation impact, availability of open-source implementation and whether the alignment task was a central focus of the study. We included only methods published no later than December 2024. To ensure fair comparisons across methods, we adhered to strict standardization protocols: (1) no additional preprocessing steps (for instance, cell and gene filtering) were applied post-data input; (2) all parameters were configured according to default settings specified in the respective method’s official documentation or tutorials; (3) for methods demonstrating applicability to multiple technological platforms datasets in their tutorials, we selected parameter sets that most closely matched each benchmark dataset’s technical specifications; and (4) because not all methods support multiple-slices (≥ 3 slices) alignment, we adopted a unified strategy. For methods capable of multislice alignment, we used their native multislice alignment function. For pairwise alignment-restricted methods, we performed pairwise alignment in an iterative manner (for example, aligning slice 2 to slice 1, then aligning slice 3 to the aligned slice 2 and so forth) and subsequently stacked all slices into a common coordinate system. Descriptions of the foundational principles of each evaluated method, together with

implementation details specific to our benchmarking framework, are provided in Supplementary Note 2.

Benchmarked datasets

Below, we detail the biological characteristics and technical specifications of real-world datasets used in our analysis. The individual tissue slice is called ‘data’, whereas collections of slices constitute ‘datasets’. All evaluated datasets were obtained from public repositories and underwent standardized preprocessing to generate uniform AnnData format inputs.

Specifically, we processed the raw data as follows: for data already in AnnData format, we removed redundant information unrelated to alignment. For data in other formats, we extracted gene expression and spatial coordinate information from each slice to create a new AnnData object, ensuring compatibility with our subsequent evaluation framework. In addition, for the computation of certain metrics, we incorporated region or cell type labels into the AnnData object. We also calculated total counts for some datasets to facilitate visualization. When reference tutorial was available for the downloaded raw data, we adhered to its processing protocols as closely as possible. Unless explicitly specified in the raw data processing tutorials, we did not perform additional quality control (for instance, filtering cells or genes) or normalization. However, we trimmed outlier points at the edges of certain datasets to ensure slices had a clean and complete shape.

Data1 to Data4. This 10x Genomics Visium ST dataset is generated from human DLPFC samples of three donors³⁶. For each individual, the dataset comprises four spatial tissue slices, labeled A, B, C and D. In all individuals, the AB and CD slice pairs are adjacent and spaced 10 μm apart, whereas the BC pair is separated by a larger 300- μm distance. Our benchmark includes the four slices from donor III, with 33,538 genes per slice and spot numbers of 3,611 (Data1), 3,635 (Data2), 3,566 (Data3) and 3,431 (Data4), respectively. The data were obtained from spatialLIBD⁵³ with region annotations ranges from layer 1 to 6 as well as the white matter. We applied random rigid transformations to all four tissue slices to perturb their original spatial orientations and positions, which were then used to evaluate the performance of various alignment methods.

Data5 to Data9. The MERFISH dataset³⁷ includes five slices of mouse hypothalamic preoptic region with annotation labels⁵⁴. These slices contain 5,488 (Data5), 5,557 (Data6), 5,926 (Data7), 5,803 (Data8) and 5,543 (Data9) cells, respectively, with a total of 155 unique genes measured per slice. The dataset was obtained from ref. 55.

Data10 to Data15. This dataset consists of ST data from four human patients diagnosed with cutaneous squamous cell carcinoma (SCC)⁵⁶. For each individual, three ST slices are available, with approximately 600–700 spots per slice. The dataset captures tumor regions, which are characterized by high spatial heterogeneity. A key confounding factor in this dataset is sequencing coverage: slices from patient 2 exhibit more than twice the sequencing coverage compared with those from patients 5, 9 and 10. In our benchmark, we selected slice sets from patient 2 and patient 5 as benchmarking datasets, designated as SCC_P2 and SCC_P5, respectively.

Data16 to Data31. This dataset was sourced from a database designed to compile 3D spatial transcriptomic profiles spanning all developmental stages of *Drosophila* embryos and larvae (Flysta3D) using the Stereo-seq platform⁴⁷. The database contains 3D ST datasets across five developmental stages, including 14–16 h and 16–18 h post egg-laying (E14h–16h and E18h–20h), as well as stages L1, L2 and L3. All the 2D slices were combined to recreate their 3D spatial transcriptomes. For our study, we selected the E14h–16h group, which contains 16 tissue slices, encompassing a total of 15,295 cells and 13,668 genes per slice. The z-coordinate and region annotations are also provided.

Data32 to Data33. This dataset was obtained from the official 10x Genomics website and consists of two consecutive sagittal–anterior mouse brain slices generated using the Visium platform⁴⁴. The first and second slices contain 2,696 and 2,825 spots, respectively, with 31,053 genes detected each slice. The cluster annotations are provided. We applied random rigid transformations to the slices to disrupt their original coordinates for subsequent evaluation of alignment methods.

Data34 to Data35. Similar to the Data32 and Data33, this one was also sourced from the 10x Genomics platform and includes two adjacent sagittal–posterior mouse brain slices⁴⁴. The slices contain 3,353 and 3,293 spots, respectively, with 31,053 genes profiled per slice. We also applied random rigid transformations to perturb the original spatial configuration for alignment benchmarking.

Data36 to Data38. This dataset is generated using the imaging-based ST platform BaristaSeq³⁰ and includes three slices derived from the mouse primary cortex. The cell numbers for each slice are 1,525, 2,042 and 1,690, with each slice profiling 79 distinct genes. The random rigid transformations were also applied as previously mentioned. This dataset is obtained from ref. 57.

Data39 to Data41. This dataset is generated using the imaging-based ST technique STARmap⁹ and also comprises three slices collected from the mouse medial prefrontal cortex, with expert-defined layer annotations available⁵⁴. The cell numbers for the three slices are 1,049 (Data39), 1,053 (Data40) and 1,088 (Data41), with each slice measuring 166 unique genes. The random rigid transformations were also applied. This dataset is obtained from ref. 57.

Data42 to Data44. This dataset originates from the official MERFISH website and includes measurements of 483 genes across three full coronal mouse brain slices from three biological replicates⁴⁵. In our study, we selected the three slices from replicate 1 (S1R1, S1R2 and S1R3). For each slice, only the right hemisphere was retained, followed by quality-control filtering. After preprocessing, the number of cells retained per slice was 40,856, 45,181 and 42,722, respectively.

Data45 to Data77. This dataset consists of 33 MERFISH slices from the mouse primary motor cortex, comprising a total of 161,384 cells and 254 genes⁵⁸. Additional region labels are provided for each slice.

Data78 to Data79. This dataset includes two cross-platform mouse olfactory bulb slices generated using Stereo-seq⁴² and Slide-seqV2³². Both slices exhibit clear tissue structures but show noticeable batch effects. The Stereo-seq slice contains 19,109 cells and 27,106 genes. For the Slide-seqV2 slice, edge trimming was applied to reduce potential artifacts, followed by quality filtering, resulting in 20,835 cells and 21,220 genes.

Data80 to Data98. This dataset was generated using the OpenST technology from the human metastatic lymph node samples³³. It includes 19 consecutive slices, totaling 1,097,769 cells and 28,943 genes. The dataset provides full 3D spatial coordinates, including the z-axis, along with annotated cell type labels.

Data99 to Data227. This dataset comprises multiple MERFISH whole mouse brain slices, with annotated region labels available⁴⁰. Slices lacking annotations—primarily small sections from the brain edges—were excluded, resulting in a final total of 129 slices. Altogether, the dataset includes 3,698,530 cells and 1,122 genes.

Data228 to Data230. This dataset includes 10x Genomics cross-platform paired ST data from human breast cancer samples, integrating one Visium slice (Data228) and two adjacent Xenium slices (Data229–230), along with corresponding histological images¹⁰. The

Visium data consists of 4,992 spots and 18,085 genes, whereas the two Xenium replicates contain 167,780 and 118,752 cells, respectively, with 313 genes profiled. All samples are accompanied by annotation labels.

Data231 to Data240. This dataset was compiled from publicly available sources, focusing specifically on mouse brain coronal sections. It comprises ten slices generated using a variety of ST platforms, including ST⁴³, 10x Visium⁴⁴, MERFISH⁴⁵, STARmap PLUS³¹, Stereo-seq⁴⁶, Slide-seq⁴¹, Xenium⁴⁴, Xenium 5k⁴⁴, Visium HD⁴⁴ and CosMx³⁴. Despite substantial variation in the number of spots or cells and the number of genes captured across platforms, all slices exhibit clear and consistent biological structures that were able to be aligned. We applied edge trimming and quality-control procedures to each slice, retaining only the right hemisphere region and converting all data into a standardized AnnData format for alignment tasks.

Evaluation metrics

For datasets without ground truth coordinates, our evaluation is primarily divided into two components: gene-based metrics and landmark-based metrics. For datasets with 3D coordinates as ground truth, we used MAE for evaluation additionally.

MAE. MAE is a commonly used metric to evaluate the average absolute difference between predicted value and ground truth value. In our context, MAE quantifies the alignment accuracy between the predicted spatial positions and the ground truth positions of corresponding spots/cells. Let $p = (x^{\text{pred}}, y^{\text{pred}})$ denote the predicted 2D coordinate and $g = (x^{\text{GT}}, y^{\text{GT}})$ represent the ground truth coordinate. Then the MAE is computed as

$$\text{MAE} = \frac{1}{2N} \sum_{i=1}^N (|x_i^{\text{pred}} - x_i^{\text{GT}}| + |y_i^{\text{pred}} - y_i^{\text{GT}}|),$$

where N is the total number of spots considered. A lower MAE value indicates that the predicted spot coordinates are closer to the true positions, reflecting higher spatial alignment accuracy. We use `mean_absolute_error` function from `scikit-learn`³⁵ to compute MAE.

Gene-based metrics

The first gene-based component is grounded in a biological assumption that is also commonly adopted in the design of many alignment methods—corresponding locations in adjacent tissue slices are more likely to exhibit similar gene expression patterns.

On the basis of this assumption, we first define the ‘overlapping area’ between two aligned slices: a rectangular area is initially designated as the overlapping region, which is then subdivided into multiple subgrids. For each subgrid, we count the number of spots (or cells) from each slice. The average number of spots in each subgrid is calculated by taking the minimum count from the two slices and then dividing it by two to determine a threshold. If either slice contains fewer spots than this threshold in a given subgrid, the subgrid is excluded. The remaining subgrids collectively constitute the ‘overlapping area’.

Within this defined overlapping area, we compute various gene-based metrics. Given a gene list—either a marker gene list reported in previous studies or one identified using the `highly_variable_genes` function in the `SCANPY` package—we calculate the average expression of each gene in the corresponding subgrids of the two slices. We then compute similarity measures, such as the PCC, between the resulting expression vectors. For datasets comprising multiple slices, we calculate pairwise similarity scores between all slice pairs and take the average as the final measure for each gene. By examining the distribution of these metrics across all marker genes, we assess the performance of alignment at the expression level.

Following this strategy, we used four different metrics. Higher similarity values indicate better correspondence between aligned positions, reflecting improved alignment accuracy.

PCC. The PCC quantifies the strength and direction of a linear correlation, ranging from -1 to 1, with 1 representing a perfect positive linear correlation and -1 indicating a complete negative one. 0 indicates that there is no linear correlation.

For two variables $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$. It is defined as

$$\text{PCC}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

where $\bar{x}$ and $\bar{y}$ are the mean values of x and y , respectively. We utilize the `pearsonr` function in SciPy package⁵⁹ for computation.

Cos_Sim. Cos_Sim measures the similarity between two vectors by calculating the cosine of the angle between them and is commonly used in high-dimensional spaces. It effectively captures the similarity in direction between vectors regardless of their magnitudes. The value ranges from -1 to 1, where -1 indicates that the vectors point in completely opposite directions, and 1 indicates perfect identical directions.

For two nonzero vectors $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$. It is defined as

$$\text{Cos_Sim}(x, y) = \frac{x \cdot y}{\|x\| \|y\|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}},$$

where $x \cdot y$ denotes the dot product of x and y , and $\|x\|$ and $\|y\|$ are the Euclidean norms of the vectors.

SSIM. The SSIM is a metric used to evaluate the structural similarity between two images x and y . SSIM considers three aspects of similarity: luminance, contrast and structure. It is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)},$$

where μ is the mean value of x and y , σ^2 is the variance, σ_{xy} is the covariance between them and C_1 and C_2 are small constants to stabilize the division when the denominators are close to zero. SSIM ranges from -1 to 1, where 1 indicates perfect structural similarity. We treat the expression values within subgrids as pixel features, such as grayscale intensity, to assess the structural similarity between tissue slices. A higher SSIM reflects greater alignment in structural features between the two slices.

MI. MI is also a commonly used metric in image registration, measuring the amount of shared information between two images. The higher the similarity or the greater the overlapping region between the images, the stronger their correlation, resulting in a higher MI value. Given two random variables X and Y , MI is defined as

$$\text{MI}(X, Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right),$$

where $p(x, y)$ is the joint probability distribution of X and Y , and $p(x)$ and $p(y)$ are their marginal probability distributions. Here, MI quantifies the shared information between the gene expression features in two slices. A higher MI value suggests a stronger statistical association and greater similarity between the two distributions. We use `mutual_info_regression` function from `scikit-learn`³⁵ to compute MI.

Landmark-based metrics

The landmark-based metrics primarily assess performance from a regional perspective. We begin by defining 'matching points' after

alignment. Once the overlapping area is identified, we use the xy -coordinates and apply the `cKDTree` function from SciPy package to find matching points: for each cell in slice 1, we identify the spatially closest cell in slice 2, thereby establishing a matching relationship. We draw inspiration from the accuracy metric used in PASTE¹⁶ with adaptations. We define the accuracy of a pairwise slice alignment as the proportion of matching point pairs that belong to the same region after alignment. Each pair of aligned spots contributes 1 if they belong to the same region and 0 otherwise.

Specifically, let S denote the set of spots in the overlap area of the reference slice, $D(i)$ denote the label (for instance, cluster or region) of spot $i \in S$ and $j(i)$ be the spot in the other slice that is considered aligned or matching to spot i .

We define the accuracy as follows:

$$\text{Accuracy} = \frac{\sum_{i \in S} \mathbf{1}\{D(i) = D(j(i))\}}{|S|},$$

where $\mathbf{1}\{D(i) = D(j(i))\}$ is an indicator function that equals 1 if the labels of spot i and spot j are the same and 0 otherwise ('1' is the symbol for an indicator function). $|S|$ is the total number of spots in the overlap region in reference slice. The closer the accuracy is to 1, the better the alignment.

Furthermore, we examine the overlap ratio between the geometric boundaries of major regions across slices following alignment. In theory, a higher overlap ratio indicates better performance. Specifically, in the overlapping area, for a given region, we use the `alpha_shapes`^{60,61} package to delineate the boundary of that region in both slices, forming 2D geometric shapes A and B . We then calculate the proportion of overlap between these shapes on the xy -plane. Let $\text{area}(A \cap B)$ denote the area of intersection of these two shapes and $\text{area}(A \cup B)$ denote the area of their union. The overlap ratio is then defined as

$$\text{Overlap ratio} = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)}.$$

We compute this overlap ratio separately for major regions and recorded each as a metric named `overlap_ratio_region`.

Finally, we compute an overall ranking score. This is done by ranking all methods on the basis of the values of each individual metric and then averaging these rankings to produce a composite score. Let N be the total number of metrics. The overall score is then calculated as follows:

$$\text{Overall ranking score} = \frac{1}{N} \sum_{n=1}^N \text{rank}(\text{metric}_n)$$

We also evaluate the consistency between metrics using Kendall's tau correlation coefficient.

Kendall's tau correlation coefficient. Kendall's tau is a rank-based correlation coefficient that measures the ordinal association between two variables x and y . A value of 1 implies perfect agreement between rankings, -1 indicates perfect disagreement and 0 implies no correlation. Kendall's tau is particularly useful for evaluating the similarity of rankings when the exact values are not important, only their order. Kendall's tau coefficient is computed as

$$\tau = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}},$$

where C is the number of concordant pairs, D is the number of discordant pairs, T_x is the number of tied pairs only in x and T_y is the number of tied pairs only in y . We computed this metric using the `cor()` function in R package⁶².

Impact of alignment on downstream-task

For downstream analysis, we aim to select a robust, widely recognized and computationally efficient method as a representative downstream tool to fairly compare different alignment approaches. We selected 3D spatial clustering as a representative task and used GraphST¹³ to perform this analysis. It is important to note that the performance of downstream tasks may be influenced by the specific clustering algorithm. Therefore, this part of the analysis serves as an initial evaluation for exploratory purposes. Future work may further explore additional downstream methods.

GraphST. GraphST is a spatial clustering method that leverages a graph self-supervised contrastive learning framework. By incorporating graph neural networks and contrastive learning strategies, it captures informative representations of ST data by jointly considering gene expression profiles and spatial positions. To handle multiple slices, GraphST first requires spatial alignment across slices. We input either the original slices or those aligned by different methods into GraphST as input for 3D clustering.

To evaluate the performance, we visualized the results and compared the labels derived from GraphST with ground truth annotations using a set of quantitative metrics from the scikit-learn package³⁵.

ARI. ARI is a statistical measure that evaluates the degree of similarity between two clustering results. It is widely adopted in benchmarking clustering methods, particularly in biological data analysis. In this study, ARI is used to assess how accurately the predicted tissue domain correspond to the true domain labels. To compute ARI, we first introduce a contingency table by comparing their clustering assignments in both the predicted and reference domains within each spot. The table includes four terms: TP for true positive, TN for true negative, FP for false positive and FN for false negative. Let E be the expected index value under random clustering. The ARI is then computed as

$$ARI = \frac{TP + TN - E}{TP + TN + FP + FN - E},$$

where E is

$$E = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{TP + TN + FP + FN}.$$

NMI. NMI is a metric to evaluate the similarity between two clustering assignments, for example, the method can use it to assess the performance of spatial domain identification⁶³. It is bounded between 0 and 1, with higher values indicating greater consistency between the clustering results. In our study, NMI is used to quantify how well the spatial domains inferred align with the ground truth region labels. Let P denote the predicted clustering and T represent the reference labels. Given that $H(P)$ and $H(T)$ are the entropies of P and T , respectively, and $MI(P, T)$ is the MI between them, NMI is calculated as

$$NMI(P, T) = \frac{MI(P, T)}{\sqrt{H(P)H(T)}}.$$

HOM. The HOM evaluates the homogeneity of a clustering result relative to a known ground truth domain label⁶⁴. A clustering is said to be perfectly homogeneous when each predicted cluster contains only samples from a single ground truth class. The HOM score ranges from 0 to 1, where a value of 1 indicates a perfectly homogeneous clustering in which each cluster corresponds entirely to one class label.

COM. The COM is a metric used to evaluate how well all members of each ground truth class are assigned to the same predicted cluster⁶⁴. A clustering solution achieves perfect completeness when all samples

that belong to the same class are grouped into a single cluster. The COM score ranges from 0 to 1. A score of 1 indicates that all data points of a class are perfectly grouped within a single cluster, reflecting complete preservation of class integrity.

Evaluation of robustness

To evaluate robustness, we designed two scenarios by altering the overlap ratio between slices and the initial orientation. We used real DLPFC slices to generate simulated data. In the first scenario, we shifted the tissue region of the target slice along the x axis (with a step size of 10% of the total tissue length along the x axis) and cropped it parallel to the y axis, retaining the left area. This process generated slices with approximately 10%, 20% and so on, up to 100% overlap with the reference slice. Each of these slices was then aligned with the reference slice to assess and record the alignment performance. In the second scenario, we rotated the target slice clockwise around its center from 0° to 360°, with a step size of 15°. Each newly generated slice was aligned with the reference slice to observe and record the robustness. In addition to visualizations, we used the aforementioned accuracy metrics for evaluation and calculated the mean and variance to demonstrate the results.

Overall performance evaluation

To evaluate the overall performance of all methods under regular scenarios, we compiled datasets that are compatible with most methods and categorized them into two types on the basis of the underlying technology: NGS-based and imaging-based datasets. The NGS-based group includes Data1–4, Data10–12, Data16–31, Data32–33 and Data34–35, whereas the imaging-based group includes Data5–9, Data34–35, Data39–41 and Data45–77. We applied the previously defined evaluation framework to each pair of aligned slices to compute quantitative metrics. The scores were then averaged across all tasks, and methods were ranked accordingly to derive a final overall ranking score. Note that if a method failed to run on a particular task, it was assigned the worst rank for that task.

Usability

We implement a method usability scale adapted from Wang et al. to assess³⁸. Our evaluation for this part includes availability, code quality, code assurance, documentation, behavior, reproducibility, paper and application. We applied different weights across all categories for overall scoring and then ranked all methods on the basis of the overall score to generate the final usability ranking (Source data). Please refer to Supplementary Table 2 for more detailed information regarding the evaluation

Strategies for large-scale data issue

For large-scale data, we propose three solutions. The first is to perform a pre-alignment step before the formal alignment. This step only requires the slices to be roughly aligned in terms of position and orientation and does not need to be highly precise. It can be carried out manually or with some very simple and fast alignment methods, depending on the user's own choice. The main purpose of this step is to bring the slices closer together and correct the initial angle, which helps reduce computation time and improve accuracy during the formal alignment.

The second approach targets methods with various parameters and detailed parameter-tuning tutorials. Generally, following the provided documentation step by step is sufficient, although it may take some time. We recommend that users first examine the characteristics of their data and refer to example tutorials that use datasets with similar features. Then, each parameter should be adjusted individually within the suggested range.

The third strategy is applicable to any dataset and focuses on reducing the data scale. To lower the resolution, we divide the dataset

into subgrids and control the resolution by adjusting the number of grids. Gene expression values within each subgrid are averaged to represent the expression profile. The center coordinate of each grid is then treated as the spot coordinate of the downsampled, lower-resolution data. This process generates a dataset with reduced resolution. Then, various alignment methods can then be applied to this low-resolution data. After alignment, we compute the transformation between the original and aligned coordinates for each slice and transfer this transformation back to the high-resolution dataset, thereby obtaining the new coordinates at full resolution—a process referred to as ‘resolution recovery’. Several checking steps are present in Supplementary Fig. 9.

When time allows, users are encouraged to experiment with multiple resolution settings. For instance, under the maximum allowable resolution for computation, users may adjust the downsampling ratio (for instance, 10%, 20% and so on) to identify the optimal resolution. It is important to note that, on the basis of our experiments, the best alignment performance does not always occur at the highest resolution within this range. Therefore, this step is essential.

Another feasible strategy to reduce the data scale is to apply conventional quality-control steps by filtering cells or genes. This step should be performed before alignment. Similarly, if time permits and the user’s primary goal is high accuracy, the optimal filtering thresholds should be determined through multiple rounds of testing.

Computational resources

Intel(R) Xeon(R) Gold 5318Y CPUs (2.10 GHz, 36 MB L3 cache, 48 cores in total), 503 GB of system memory and an NVIDIA A100 80GB PCIe graphics processing unit (GPU) with 80 GB of GPU memory (CUDA version 12.2) were used.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this Article.

Data availability

Data1 to Data4 were downloaded from ref. 53 (<https://doi.org/10.1186/s12864-022-08601-w>). Data5 to Data9 were obtained from ref. 55 (<https://doi.org/10.5061/dryad.8t8s248>). Data10 to Data15 were downloaded from ref. 56 (<https://doi.org/10.1016/j.cell.2020.08.043>). Data16 to Data31 were obtained from <https://db.cngb.org/stomics/flysta3d/>. Data32 to Data33 were obtained from <https://www.10xgenomics.com/cn/datasets/mouse-brain-serial-section-1-sagittal-anterior-1-standard-1-0-0> and <https://www.10xgenomics.com/cn/datasets/mouse-brain-serial-section-2-sagittal-anterior-1-standard>. Data 34 to Data35 were obtained from <https://www.10xgenomics.com/cn/datasets/mouse-brain-serial-section-1-sagittal-posterior-1-standard-1-1-0> and <https://www.10xgenomics.com/cn/datasets/mouse-brain-serial-section-2-sagittal-posterior-1-standard>. Data36 to Data38 were obtained from ref. 57. Data39 to Data41 were obtained from ref. 57 (<https://doi.org/10.1038/s41592-024-02215-8>). Data42 to Data44 were obtained from <https://info.vizgen.com/mouse-brain-data>. Data45 to Data77 were obtained from ref. 58 (<https://doi.org/10.1038/s41586-021-03705-x>). Data78 to Data79 were obtained via GitHub at https://github.com/JinmiaoChen-Lab/SEDR_analyses and at https://singlecell.broadinstitute.org/single_cell/study/SCP815, respectively. Data80 to Data98 were obtained from ref. 33 (<https://doi.org/10.1016/j.cell.2024.05.055>). Data99 to Data227 were obtained from ref. 40 (<https://doi.org/10.1038/s41586-023-06808-9>). Data228 to Data230 were obtained from ref. 10 (<https://doi.org/10.1038/s41467-023-43458-x>). Data231 to Data240 were downloaded via ST at <https://www.molecularatlas.org/st-js-viewer> (21A, ref. 43), via 10x Visium at <https://www.10xgenomics.com/datasets/adult-mouse-brain-coronal-section-fresh-frozen-1-standard> (ref. 44), via MERFISH at <https://info.vizgen.com/mouse-brain-data> (S2R3, ref. 45), STARmap PLUS via Zenodo at <https://doi.org/10.5281/zenodo.8327576> (well11, ref. 31), via Stereo-seq at <https://mouse.digital-brain.cn/spatial-omics> (T312, ref. 46),

via Slide-seq at <https://www.braincelldata.org/cellspace> (46, ref. 41), via Xenium at <https://www.10xgenomics.com/datasets/fresh-frozen-mouse-brain-replicates-1-standard> (ref. 44), via Xenium 5k at <https://www.10xgenomics.com/datasets/xenium-prime-fresh-frozen-mouse-brain> (ref. 44), via Visium HD at <https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-mouse-brain-fresh-frozen> (ref. 44) and via CosMx at <https://nanosttring.com/products/cosmx-spatial-molecular-imager/ffpe-dataset/cosmx-smi-mouse-brain-ffpe-dataset/> (ref. 34). Source data are provided with this paper.

Code availability

The code and tutorials are available via GitHub at <https://github.com/Yunzhi-Yan/SABench> and via Zenodo at <https://doi.org/10.5281/zenodo.18605715> (ref. 65).

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Author contributions

Author contributions are described using CRediT roles (<https://credit.niso.org/>). Y.Y.: conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing—original draft and writing—review and editing. T.G.: investigation, methodology, software and writing—review and editing. C.S.: investigation, methodology, software and writing—review and editing. Y. Zhang: investigation, methodology, software and writing—review and editing. Y.C.: investigation, methodology and software. S. Lin: software. Q.Z.: software. Y.D.: investigation. C.H.: investigation. K.K.: investigation. S. Li: investigation. Y. Zhao: investigation. Z.L.: investigation. Z.Y.: conceptualization, investigation, methodology, resources, supervision and writing—review and editing. B.-Z.Q.: conceptualization, investigation, methodology, resources, supervision and writing—review and editing.

Competing interests

The authors declare no competing interests.

Additional information

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Data collection	No data was generated. Public data were downloaded from both public repositories
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 Data45 to Data77 were obtained from ref58.
 Data78 to Data79 were obtained from https://github.com/JinmiaoChenLab/SEDR_analyses and https://singlecell.broadinstitute.org/single_cell/study/SCP815, respectively. Data80 to Data98 were obtained from ref33.
 Data99 to Data227 were obtained from ref40.
 Data228 to Data230 were obtained from ref10.
 Data231 to Data240 were downloaded from ST ref43(<https://www.molecularatlas.org/st-js-viewer>, 21A), 10x Visium ref44(<https://www.10xgenomics.com/datasets/adult-mouse-brain-coronal-section-fresh-frozen-1-standard>), MERFISH ref45(<https://info.vizgen.com/mouse-brain-data>, S2R3), STARmap PLUS ref31(<https://zenodo.org/records/8327576>, well11), Stereo-seq ref46(<https://mouse.digital-brain.cn/spatial-omics>, T312), Slide-seq ref41(<https://www.braincelldata.org/cellspace>, 46), Xenium ref44 (<https://www.10xgenomics.com/datasets/fresh-frozen-mouse-brain-replicates-1-standard>), Xenium 5k ref44 (<https://www.10xgenomics.com/datasets/xenium-prime-fresh-frozen-mouse-brain>), Visium HD ref44 (<https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-mouse-brain-fresh-frozen>), and CosMx ref34(<https://nanosttring.com/products/cosmx-spatial-molecular-imager/ffpe-dataset/cosmx-smi-mouse-brain-ffpe-dataset/>).

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Methods

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